CardSense

Real-Time Card Game Assistant using Computer Vision

Deep Learning — 6th Semester | Rana Hammad (475203) • Abdul Wahab (464018) • Abu Bakar (455037)

1. What Is This Project?

CardSense is a phone-camera-based assistant that helps users play card games. The app uses deep learning to detect and track cards in real time. It then gives the user simple suggestions based on the game state.

Supported games: Bluff, Thulla (Mendikot), and Teen Patti. More games can be added later.

This is inspired by wearable AI cameras (like Meta glasses) but built on a phone to keep things practical for a course project.

2. The Problem

Card games require players to remember a lot — what cards were played, what opponents claimed, and what cards are still in the game. This is hard to track manually, especially in fast-paced rounds.

CardSense solves this by watching the game through the camera and doing the tracking automatically.

3. Objectives

- Train a model to detect and classify playing cards in a live video feed.
- Track which cards have been played and by whom.
- Build rule engines for Bluff, Thulla, and Teen Patti.
- Show suggestions to the user through a simple mobile UI.

4. How It Works

Step	What Happens	Technology
1. Capture	Phone camera records the game	Android/iOS camera
2. Detect	Cards are found in the frame	YOLOv8 (fine-tuned)
3. Classify	Each card's rank and suit is identified	CNN (EfficientNet-B0)
4. Track	Cards are tracked across frames	DeepSORT / ByteTrack
5. State	Game state is updated each round	Custom logic module
6. Suggest	Tips shown to user based on game rules	Rule engine per game

The table is divided into player zones by seating position. Cards detected in each zone are assigned to that player.

For Bluff — where cards are placed face-down — the system uses memory of previously seen cards and probability to make inferences.

5. Dataset

- Primary: Roboflow playing card dataset (~5,000 annotated images, 52 classes).
- Custom: We will film real card games and annotate them ourselves to capture local conditions (lighting, card designs, etc.).
- Augmentation: Rotation, blur, occlusion, and brightness shifts using Albumentations.

Target dataset size: ~8,000–10,000 images.

6. Timeline

Weeks	Task
1–2	Dataset collection, literature review, environment setup
3–4	Train YOLOv8 baseline model; collect custom data and fine-tune
5–6	Card tracking (DeepSORT) and game state module
7–8	Rule engines for Bluff, Thulla, and Teen Patti
9–10	Mobile app integration, testing, evaluation, and final demo

7. Team

Member	Role	Responsibilities
Rana Hammad (475203)	ML Engineer	Model training, dataset prep, evaluation
Abdul Wahab (464018)	Backend	Game state module, rule engines, Bluff probability logic
Abu Bakar (455037)	Mobile Dev	App development, camera integration, UI

8. How We'll Measure Success

- Card detection mAP@0.5 ≥ 0.85 on test set.
- Card classification accuracy ≥ 92%.
- Real-time speed ≥ 15 FPS on a mid-range phone.
- Game state accuracy checked against ground truth across 10 recorded sessions.

9. Tools & Stack

Area	Tools
Model Training	PyTorch, YOLOv8 (Ultralytics)
Tracking	DeepSORT / ByteTrack
Annotation	Roboflow, LabelImg
Mobile App	Flutter or React Native
On-device Inference	TensorFlow Lite / ONNX Runtime
Experiment Tracking	Weights & Biases (wandb)

10. Known Challenges

Challenge	How We Handle It
Cards partially hidden in hand	Train with occluded card images; use tracking to fill gaps
Bluff cards are face-down	Use memory + probability — not direct detection
Running fast enough on mobile	Use quantized YOLOv8-nano (INT8) via TFLite
Different lighting conditions	Heavy augmentation during training

11. Ethical Note

CardSense is an educational and accessibility tool. It helps users — especially beginners or those with cognitive difficulties — keep track of the game. It does not reveal hidden opponent cards. It only uses information the user is entitled to see.